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Combined Effects of Static and Dynamic Stretching on the Muscle-Tendon Unit Stiffness and Strength of the Hamstrings

Kosuke Takeuchi,[1] Masatoshi Nakamura,[2] Shingo Matsuo,[3] Mina Samukawa,[4] Taichi Yamaguchi,[5] and Takamasa Mizuno[6]

[1]Department of Physical Therapy, Kobe International University, Kobe-shi, Japan; [2]Faculty of Rehabilitation Sciences, Nishi Kyushu University, Kanzaki-cho, Japan; [3]Department of Rehabilitation, Faculty of Health Sciences, Nihon Fukushi University, Handa-shi, Japan; [4]Faculty of Health Sciences, Hokkaido University, Kita-ku, Japan; [5]Laboratory of Food Ecology and Sports Science, Department of Foods Science and Human Wellness, College of Agriculture, Food and Environment Science, Rakuno Gakuen University, Ebetsu, Japan; and [6]Research Center of Health, Physical Fitness and Sports, Nagoya University, Nagoya-shi, Japan

Abstract

Takeuchi, K, Nakamura, M, Matsuo, S, Samukawa, M, Yamaguchi, T, and Mizuno, T. Combined effects of static and dynamic stretching on the muscle-tendon unit stiffness and strength of the hamstrings. *J Strength Cond Res* 38(4): 681–686, 2024—Combined static and dynamic stretching for 30 seconds is frequently used as a part of a warm-up program. However, a stretching method that can both decrease muscle-tendon unit (MTU) stiffness and increase muscle strength has not been developed. The purpose of this study was to examine the combined effects of 30 seconds of static stretching at different intensities (normal-intensity static stretching [NS] and high-intensity static [HS]) and dynamic stretching at different speeds (low-speed dynamic [LD] and high-speed dynamic stretching [HD]) on the MTU stiffness and muscle strength of the hamstrings. Thirteen healthy subjects (9 men and 4 women, 20.9 ± 0.8 years, 169.3 ± 7.2 cm, 61.1 ± 8.2 kg) performed 4 types of interventions (HS-HD, HS-LD, NS-HD, and NS-LD). Range of motion (ROM), passive torque, MTU stiffness, and muscle strength were measured before and immediately after interventions by using an isokinetic dynamometer machine. In all interventions, the ROM and passive torque significantly increased ($p < 0.01$). Muscle-tendon unit stiffness significantly decreased in HS-HD and HS-LD (both $p < 0.01$), but there was no significant change in NS-HD ($p = 0.30$) or NS-LD ($p = 0.42$). Muscle strength significantly increased after HS-HD ($p = 0.02$) and NS-LD ($p = 0.03$), but there was no significant change in HS-LD ($p = 0.23$) or NS-LD ($p = 0.26$). The results indicated that using a combination of 30 seconds of high-intensity static stretching and high-speed dynamic stretching can be beneficial for the MTU stiffness and muscle strength of the hamstrings.

Key Words: range of motion, peak torque, passive torque, short duration, warm-up routine

Introduction

Short-duration static and dynamic stretching, typically lasting 20 and 30 seconds, are frequently used as a part of a warm-up routine to change mechanical properties such as muscle-tendon unit (MTU) stiffness and increase muscle strength (11,34). Muscle-tendon unit stiffness is calculated from the torque-angle curve during passive joint movement, and alteration of the stiffness indicates an alteration in the passive elastic properties of the MTU (3). Previous studies have demonstrated that excessive stiffness can result in lower-body injuries occurring in noncontact situations (24,40) because of a diminished cushioning effect from soft tissues (6), and reducing MTU stiffness is important to prevent sports-related injuries. However, both 20 seconds of static stretching (31) and 30 seconds of dynamic stretching (35) have negligible effects on the MTU stiffness of the hamstrings. Furthermore, static stretching for more than 30 seconds has a significant negative effect on muscle strength (4), mainly due to a reduction in muscle activity (5,38,39) rather than changes in MTU stiffness (5,23). Moreover, dynamic stretching for more than 90 seconds is needed to increase muscle strength (4). Therefore, it is suggested that static and dynamic stretching for the short duration used in warm-up programs do not achieve both decreasing MTU stiffness and increasing muscle strength, and it is necessary to establish stretching methods that can achieve both these goals.

An intervention that increases muscle activity can improve a stretching-induced muscle strength deficit resulting from static stretching (33,36). Takeuchi et al. reported that 5 minutes of static stretching decreased the muscle strength of the ankle plantar flexors through a decrease in muscle activity, but performing aerobic exercise after static stretching improved the deficits in muscle strength and activity induced by stretching (33,36). In addition, dynamic stretching increases muscle activity through postactivation potentiation enhancement of the muscle (8,21), which is a transient improvement of muscle contractile performance after previous contractile activities (21,26). Previous studies reported that high-speed dynamic stretching increased muscle activity compared with low-speed stretching (8,21) and improved the hamstrings muscle strength even with 30 seconds of stretching duration (35). Therefore, it is suggested that high-speed dynamic stretching after static stretching could increase muscle strength, even if a stretching-induced muscle strength deficit occurs.

Address correspondence to Kosuke Takeuchi, ktakeuchi@kobe-kiu.ac.jp.

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A change in the MTU stiffness of the hamstrings is related to the total load of stretching (28,30). The total load is affected by stretching intensity and duration, and longer duration (15,19,28,30) or higher intensity (12,27–29,31,32) static stretching leads to a greater reduction in the MTU stiffness of the hamstrings. The intensity of static stretching is determined based on the range of motion (ROM) of each subject, and static stretching at 100%ROM is described as the normal intensity (28,31,32). It was reported that the MTU stiffness of the hamstrings decreased after 180 seconds of static stretching at 100% ROM (15,19) but within 20 seconds at 120%ROM (31,32). On the other hand, 30 seconds of dynamic stretching does not change the MTU stiffness of the hamstrings regardless of its speed and amplitude (35). There is no previous study that examined the combined effects of static and dynamic stretching on the MTU stiffness of the hamstrings. However, using a combination of static and dynamic stretching may cause a greater change in MTU stiffness than each type of stretching alone because of an increase in the total stretching load.

Therefore, the purpose of this study was to examine the combined effects of 30 seconds of static stretching at different intensities (normal-intensity and high-intensity) and dynamic stretching at different speeds (low-speed and high-speed) on the MTU stiffness and muscle strength of the hamstrings. It was hypothesized that combined high-intensity static stretching and high-speed dynamic stretching decreases the MTU stiffness and increases muscle strength of the hamstrings based on previous studies (31,32,35).

Methods

Experimental Approach to the Problem

In this study, a randomized repeated-measures experimental design was used. The subjects underwent 4 different stretching interventions with an interval of ≧24 hours between visits, in a random order (Figure 1). Subjects attended a familiarization session ≧24 hours before the first testing day. In the familiarization session, subjects experienced static and dynamic stretching interventions, and muscle strength and flexibility measurements. The data collection was as follows: rest for 5 minutes (no warm-up), preflexibility measurement, premuscle strength measurement, static stretching, dynamic stretching, postflexibility measurement, and postmuscle strength measurement. To assess changes in the flexibility of the hamstrings in the dominant limb (ball kicking preference) (28), ROM, passive torque at the maximum knee extension angle, and MTU stiffness were measured before and immediately after stretching interventions. The experiment was performed in a university laboratory, where the temperature was maintained at 25° C.

Subjects

Thirteen healthy, recreationally active subjects (9 men, 20.9 ± 0.8 years, 171.4 ± 7.3 cm, 63.5 ± 8.2 kg; 4 women, 21.0 ± 0.8 years, 163.5 ± 2.5 cm, 54.5 ± 3.3 kg) were recruited. The inclusion criteria were that subjects did not regularly perform any flexibility and strength training and had no history of lower-limb pathology. The sample size of the MTU stiffness and muscle strength was calculated with a power of 80%, alpha error of 0.05, and effect size *f* of 0.25 (middle) (31,35) using G*Power 3.1 software (Heinrich Heine University, Düsseldorf, Germany), and the results showed that the requisite number of subjects for this study was 12 subjects; thus, 13 subjects were recruited to account for possible attrition. All subjects were informed of the requirements and risks associated with their involvement in this study and signed a written informed consent document. The study was performed in accordance with the Declaration of Helsinki (1964). The Ethics Committee of Kobe International University approved the study (No. G2022-170).

Sitting Position of Measurements

The flexibility assessment was performed in the same fashion as previous studies (27–29,32). An isokinetic dynamometer machine (CYBEX NORM, Humac, CA) was used in this study. The subjects sat on the dynamometer with the backrest upright. A wedge-shaped wooden frame of 30° was inserted between the backrest and the trunk. The trunk and thigh of the dominant leg were firmly secured with straps. The knee joint was aligned with the axis of the rotation of the isokinetic dynamometer machine. The lever arm attachment was placed just proximal to the malleolus medialis and stabilized with straps. In this study, knee joint angles were reported using the isokinetic dynamometer machine. A 90-degree angle between the lever arm and floor was defined as 0 degrees of knee flexion or extension. The subjects were instructed to relax during the flexibility assessment.

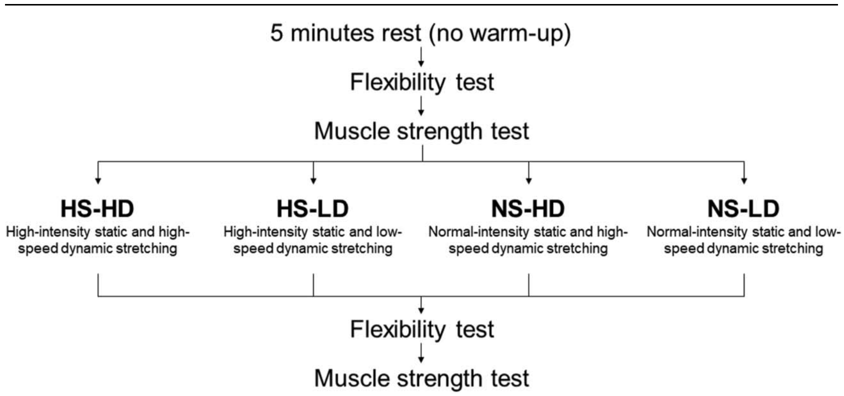

Figure 1. Experimental design. HS = high-intensity static stretching (120% range of motion); NS = normal-intensity static stretching (100% range of motion); HD = high-speed dynamic stretching (60 rpm); LD = low-speed dynamic stretching (30 rpm).

Range of Motion, Passive Torque, and Muscle-Tendon Unit Stiffness

The knee joint was passively extended from 0° to the maximum angle just before pain at $5° \cdot s^{-1}$. A previous study showed that this velocity does not cause a stretch reflex (18). Range of motion was defined as the range from 0° to the maximum knee extension angle. Passive torque and knee extension angles during ROM measurement were recorded by the isokinetic dynamometer, and the data were later exported to a computer. Passive torque at the maximum knee extension angle was used for further analysis.

Muscle-tendon unit stiffness was calculated from the same measurement points before and after stretching interventions. Muscle-tendon unit stiffness was calculated as the slope of the second-order polynomial torque-angle regression curve of the 5 measurement points (80, 85, 90, 95, and 100%ROM) measured before stretching interventions (16,17,25). However, if the maximum knee extension angle measured after stretching was smaller than that before stretching, MTU stiffness was calculated based on the ROM of postmeasurement (12,28,32).

Muscle Strength

The muscle strength measurement was conducted in the same fashion as a previous study (31,35). The peak torque of knee flexion during maximum voluntary isokinetic concentric contraction at $60° \cdot s^{-1}$ was measured. The subjects were secured on the isokinetic dynamometer machine in the same fashion as the ROM assessment. The range of movement was set from 0° to the maximum knee extension angle. The subjects performed 3 submaximal trials on the isokinetic dynamometer machine. After the submaximal trials, the subjects performed 3 maximum voluntary isokinetic concentric contractions. The greatest value of the 3 repetitions was used for the analyses.

Stretching Interventions

The subjects were secured on the isokinetic dynamometer machine in the same fashion as the ROM measurement (28,31). Static stretching for 30 seconds was performed at 2 different intensities based on the ROM of each subject (HS and NS). At NS-HD and NS-LD, the intensity of static stretching was set at the knee angle of the ROM. At HS-HD and HS-LD, the intensity of static stretching was set at the knee angle of 1.2 times the ROM (i.e., 120%ROM). The subjects were instructed to relax during static stretching.

Immediately after static stretching, 30 seconds of dynamic stretching was performed at 2 different speeds (HD and LD) (35). Subjects stood upright with each hand grasping a horizontal bar at waist height. Then, subjects flexed their hip joint with their knee joint extended at a tempo dictated by a metronome so that their dominant leg swung up to their anterior aspect and their hamstrings were stretched. In NS-LD and HS-LD, dynamic stretching was performed at 30 rpm (15 repetitions for 30 seconds). In NS-HD and HS-HD, dynamic stretching was performed at 60 rpm (30 repetitions for 30 seconds). Dynamic stretching was performed at the hip flexion range just before any pain in the hamstrings, the range which was confirmed before the stretching intervention by slowly flexing the hip joint to the maximum angle. During dynamic stretching, 2 examiners confirmed that no trunk or knee joint compensations occurred.

Reliability

The test-retest reliability for all dependent variables was determined in 7 subjects. The 2 tests were separated and performed on 3 different days at the same time of the day. The reliability of ROM (intraclass correlation coefficient [ICC] of 0.95), passive torque at the maximum knee extension angle (ICC of 0.89), MTU stiffness of 5 measurement points (ICC of 0.80–0.92), and muscle strength (ICC of 0.97) were acceptable in this study.

Statistical Analysis

The distribution of the data was assessed using the Shapiro-Wilk test, and it was confirmed that the data followed a normal distribution. All variables were described as mean ± *SD*. For ROM, passive torque at the maximum knee extension angle, and muscle strength data, a 2-way repeated analysis of variance (ANOVA; time [pre vs. post] × interventions [HS-HD vs. HS-LD vs. NS-HD vs. NS-LD]) was used to analyze the interaction and main effect. A repeated 3-way ANOVA (time [pre vs. post] × angle [80%ROM vs. 85%ROM vs. 90%ROM vs. 95%ROM vs. 100%ROM] × interventions [HS-HD vs. HS-LD vs. NS-HD vs. NS-LD]) was used to analyze the MTU stiffness data. If a significance was detected, post hoc analyses using Bonferroni's test were performed to determine where significant differences occurred. Partial eta-squared values were reported to reflect the magnitude of the differences for each treatment (small = 0.01, medium = 0.06, and large = 0.14) (7). Cohen's d was calculated as the mean difference between pre and post values, divided by the pooled *SD* between pre and post (7). A d of 0.00–0.19 was considered trivial, 0.20–0.49 was considered small, 0.50–0.79 was considered moderate, and ≥0.80 was considered large. The analyses were performed using SPSS version 29 (SPSS, Inc., Chicago, IL). Differences were considered statistically significant at an alpha level of $p < 0.05$.

Results

Range of Motion

There was no significant interaction ($p = 0.08$, partial eta squared = 0.17) and no main effect for interventions ($p = 0.86$, partial eta squared = 0.02), but there was a significant main effect for time ($p < 0.01$, partial eta squared = 0.84; Figure 2). Range of motion significantly increased ($p < 0.01$) after stretching interventions (HS-HD, $d = 1.29$, 95% confidence interval [CI] of 3.68–10.12; HS-LD, $d = 1.15$, 95% CI of 3.68–11.75; NS-HD, $d = 0.83$, 95% CI of 0.71–4.31; NS-LD; $d = 1.07$, 95% CI of 2.45–8.81).

Passive Torque at the Maximum Knee Extension Angle

There was no significant interaction ($p = 0.27$, partial eta squared = 0.10) and no main effect for interventions ($p = 0.11$, partial eta squared = 0.15), but there was a significant main effect for time ($p < 0.01$, partial eta squared = 0.57; Figure 2). Passive torque significantly increased ($p < 0.01$) after stretching interventions (HS-HD, $d = 0.82$, 95% CI of 0.72–4.78; HS-LD, $d = 0.74$, 95% CI of 0.68–7.04; NS-HD, $d = 0.76$, 95% CI of 0.34–2.99; NS-LD, $d = 0.98$, 95% CI of 0.79–3.35).

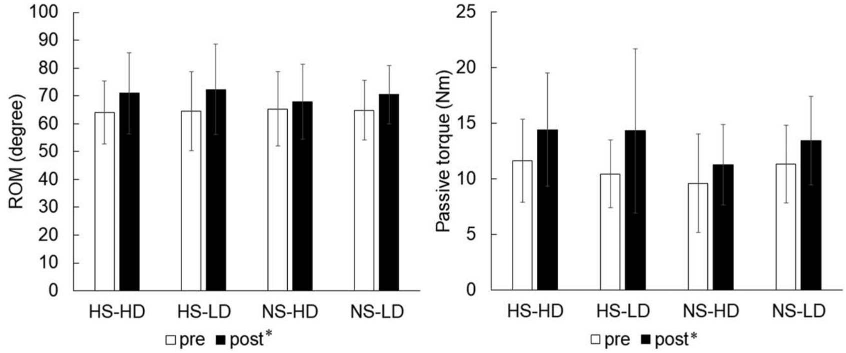

Figure 2. Changes in range of motion and passive torque at the maximum knee extension angle. Data were represented as mean ± *SD*. $^{*}p < 0.01$ vs. pre-value. ROM = range of motion; HS = high-intensity static stretching; NS = normal-intensity static stretching; HD = high-speed dynamic stretching; LD = low-speed dynamic stretching.

Muscle-Tendon Unit Stiffness

There was no significant 3-way interaction between time, interventions, and angle ($p = 0.91$, partial eta squared = 0.04; Table 1). There were no significant 2-way interactions between interventions and angle ($p = 0.93$, partial eta squared = 0.03) or time and angle ($p = 0.91$, partial eta squared = 0.01), but there was a significant 2-way interaction between time and interventions ($p < 0.01$, partial eta squared = 0.38). Muscle-tendon unit stiffness decreased in HS-HD ($p < 0.01$, 95% CI of −0.06 to −0.25) and HS-LD ($p < 0.01$, 95% CI of −0.07 to −0.19), but there was no significant change in NS-HD ($p = 0.30$, 95% CI of −0.05 to 0.14) and NS-LD ($p = 0.42$, 95% CI of −0.06 to 0.13).

Muscle Strength

There was a significant 2-way interaction ($p = 0.03$, partial eta squared = 0.22; Figure 3). Muscle strength significantly increased in HS-HD ($p = 0.02$, $d = 0.77$, 95% CI of 0.80–6.54) and NS-HD ($p = 0.03$, $d = 0.50$, 95% CI of 0.36–4.98), but it was not significantly changed in HS-LD ($p = 0.23$, $d = -0.35$, 95% CI of −9.30 to 2.45) and NS-LD ($p = 0.26$, $d = -0.32$, 95% CI of −3.58 to 1.07).

Discussion

Static and dynamic stretching are frequently used to decrease MTU stiffness and increase muscle strength as a part of a warm-up routine. However, no stretching interventions that can both decrease MTU stiffness and increase muscle strength have been developed in previous studies. This study examined the combined effects of static stretching at different intensities and dynamic stretching at different speeds on the MTU stiffness and muscle strength of the hamstrings. The results of this study found that combined high-intensity static stretching and high-speed dynamic stretching can decrease the MTU stiffness and increase the muscle strength of the hamstrings.

In this study, the MTU stiffness of the hamstrings significantly decreased in high-intensity static stretching (HS-HD and HS-LD), although it was not changed in normal-intensity static stretching (NS-HD and NS-LD). Takeuchi et al. reported a significant negative relationship between stretching load and relative change in the MTU stiffness of the hamstrings (28,30). The stretching load is mainly affected by stretching intensity and duration (28,30), and longer duration (15,19,28,30) or higher intensity (28,30–32) static stretching has a greater effect on change in the MTU stiffness of the hamstrings. In static stretching at 120%ROM, a decrease in the MTU stiffness of the hamstrings reaches a plateau within 20 seconds (32). Therefore, it was indicated that the decrease in the MTU stiffness of the hamstrings plateaued within 30 seconds of high-intensity static stretching and that the change did not disappear with subsequent dynamic stretching. On the other hand, in static stretching at 100%ROM, 180 seconds of stretching duration is needed to decrease the MTU stiffness of the hamstrings (15,19). In addition, 30 seconds of dynamic stretching does not decrease the MTU stiffness of the hamstrings regardless of its speed and amplitude (35). There is no previous study that examined the combined effect of static and dynamic stretching on

Table 1.
Changes in muscle-tendon unit stiffness.*†

		80%ROM	85%ROM	90%ROM	95%ROM	100%ROM
HS-HD	Pre	0.38 ± 0.16	0.46 ± 0.17	0.54 ± 0.19	0.63 ± 0.22	0.71 ± 0.25
	Post‡	0.20 ± 0.24	0.29 ± 0.22	0.39 ± 0.22	0.48 ± 0.23	0.58 ± 0.26
HS-LD	Pre	0.34 ± 0.14	0.43 ± 0.13	0.52 ± 0.13	0.61 ± 0.14	0.70 ± 0.17
	post‡	0.24 ± 0.13	0.32 ± 0.13	0.39 ± 0.14	0.47 ± 0.17	0.54 ± 0.20
NS-HD	Pre	0.24 ± 0.31	0.34 ± 0.23	0.44 ± 0.20	0.54 ± 0.24	0.64 ± 0.31
	Post	0.28 ± 0.15	0.38 ± 0.15	0.49 ± 0.19	0.59 ± 0.25	0.70 ± 0.32
NS-LD	Pre	0.24 ± 0.44	0.35 ± 0.28	0.46 ± 0.15	0.57 ± 0.17	0.67 ± 0.32
	Post	0.34 ± 0.16	0.41 ± 0.18	0.49 ± 0.22	0.57 ± 0.27	0.65 ± 0.31

*ROM = range of motion; HS = high-intensity static stretching; NS = normal-intensity static stretching; HD = high-speed dynamic stretching; LD = low-speed dynamic stretching.
†Data were represented as mean ± *SD*.
‡$p < 0.01$ vs. pre-value in the same intervention.

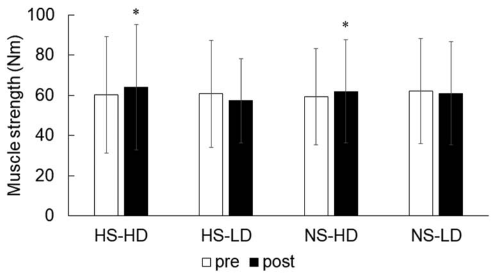

Figure 3. Change in muscle strength of the hamstrings. $^*p < 0.05$ vs. pre-value in the same intervention. HS = high-intensity static stretching; NS = normal-intensity static stretching; HD = high-speed dynamic stretching; LD = low-speed dynamic stretching.

MTU stiffness. However, the results of this study suggested that using a combination of 30 seconds of static stretching at 100% ROM and dynamic stretching was not enough to decrease the MTU stiffness of the hamstrings.

Alteration in ROM after stretching intervention is mainly affected by alteration in stretching tolerance and MTU stiffness of the muscle (3). In this study, the passive torque at the maximum knee extension angle was measured to evaluate stretching tolerance (3,14). The results of this study showed that ROM and passive torque at the maximum knee extension angle increased after all stretching interventions without any difference between interventions, but the MTU stiffness decreased in HS-HD and HS-LD. The reasons for this discrepancy were not clear from the results of this study. However, other factors, such as connective tissue (10,20), nerves (2,37), stress relaxation (9), etc., may be involved.

In this study, the muscle strength of the hamstrings significantly increased after high-speed dynamic stretching (HS-HD and NS-HD), but it was not changed after low-speed dynamic stretching (HS-LD and NS-LD). The magnitude of differences in muscle strength in HS-HD ($d = 0.77$) and NS-HD ($d = 0.50$) was both moderate and which is the same magnitude as the previous study examining the effects of 30 seconds of high-speed dynamic stretching alone ($d = 0.79$) (35). High-speed dynamic stretching increases muscle activity compared with low-speed stretching through postactivation potentiation enhancement (8,21). Moreover, it was reported that dynamic stretching could nullify the sprint performance deficit after static stretching in soccer players (1) and elite Gaelic footballers (13). These results indicated that 30 seconds of high-speed dynamic stretching could increase the muscle strength of the hamstrings regardless of the previous static stretching intensities. A previous study reported that muscle strength decreased after static stretching for more than 30 seconds (4). In this study, static stretching for 30 seconds was performed and no decrement in muscle strength was observed in any of the conditions. These results suggested that even low-speed dynamic stretching may have contributed to the maintenance of muscle strength.

There was no previous study that examined the combined effects of static and dynamic stretching on MTU stiffness and muscle strength. Therefore, it is not possible to compare the results of this study and those of the previous studies. Moreover, the results examined the effects of static and dynamic stretching in recreationally active young men and women, and different cohorts may lead to different results.

This study examined the combined effects of static stretching at different intensities and dynamic stretching at different speeds on MTU stiffness and muscle strength of the hamstrings. The results showed that MTU stiffness decreased after high-intensity static stretching (HS-HD and HS-LD), whereas muscle strength increased after high-speed dynamic stretching (HS-HD and NS-HD). Therefore, using a combination of high-intensity static stretching and high-speed dynamic stretching can both decrease the MTU stiffness and increase the muscle strength of the hamstrings.

Practical Applications

Static and dynamic stretching are frequently used in a warm-up program to decrease MTU stiffness and increase muscle strength. The hamstrings are the most common site of muscle strain (22), and many athletes need to decrease the MTU stiffness of the hamstrings to prevent muscle strain. Static stretching is useful for decreasing the MTU stiffness of the hamstrings (31), whereas dynamic stretching is useful for increasing muscle strength (35). Previous studies reported that static stretching for 180 seconds (15,19) and dynamic stretching for 90 seconds (4) were required to decrease the MTU stiffness and strength of the hamstrings, respectively. However, those stretching durations are long for many athletes and can be difficult to sustain. The results of this study indicated that combined high-intensity static stretching and high-speed dynamic stretching significantly decreased the MTU stiffness and increased the muscle strength of the hamstrings, even if the stretching duration was only 30 seconds. Therefore, it was suggested that the stretching method was ideal in that it could both prevent muscle strain and improve performance in athletes.

Acknowledgments

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